

Embedded Software Engineering

Project Name: Home Security System

Written by Joshua Anto



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Project Overview

In this project, we will use a Raspberry Pi equipped with a SensorHat to capture data from various sensors and then transfer that data to a remote database for more efficient processing. The idea behind this is to emulate on how we can implement the use of a raspberry pi and sensor hat to operate as a home security system.

Data will be collected from the accelerometer, temperature sensor, humidity sensor and pressure sensor. Data will then be sent off to PhpMyAdmin to remote store the data and provide access to users.

Use Cases

A use case for this system, imagine you are a homeowner who wants to enhance the security of your property. You've set up a Raspberry Pi-based home security system using a Sensor HAT as described. Here's how your system would work in a real-life scenario.

Motion Detection

The motion sensor collects data from the accelerometer which is connected to an external detector and placed in a key area of your home. It detects any movements or vibrations, especially when you are not at home or during the night.

Sensor Data Collection

The system continuously collects data from multiple sensors,

- Accelerometer (Motion Sensor) detects any unauthorized entry or movement.
- Temperature Sensor Monitors temperature to ensure it stays within a comfortable range.
- Humidity Sensor Measures humidity levels to prevent damage from excessive moisture.
- Pressure Sensor can be connected to a pressure plate to indicate intrusion or track barometric pressure changes to indicate open doors or windows.

Alerting and Monitoring

Motion detection triggers an alert for potential intrusions. Temperature, humidity, and pressure data are checked for unusual changes, which can also trigger alerts.

Remote Access

You and authorized users can access the system remotely through a web interface, allowing you to monitor real-time sensor data.

Notifications

Alerts are sent as notifications to you and other authorized users via email, text messages, and mobile app alerts.

Response

Depending on the alert type, actions can be taken, such as dispatching security personnel for motion alerts or checking for open doors/windows for environmental alerts.

Remote Control

You can remotely arm or disarm the system and access security camera feeds for live monitoring.

Functional Requirements

1. The system must accurately detect motion in the designated area, it should trigger an alert when motion is detected.
2. The system should collect data from the accelerometer, temperature, humidity, and pressure sensors and collected at regular intervals.
3. Data collected should then be sent to the remote database, PhpMyAdmin.

Non-Functional Requirements

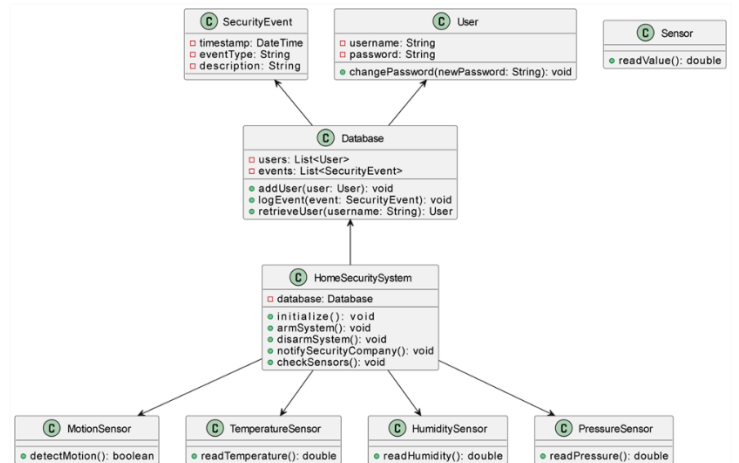
1. Data Transmission should be secure to prevent unauthorized access.
2. The system should be highly reliable, ensuring minimal false alarms and downtimes.
3. Users should find it easy to arm and disarm the system remotely.

Architectural Design of Software

UML Diagrams

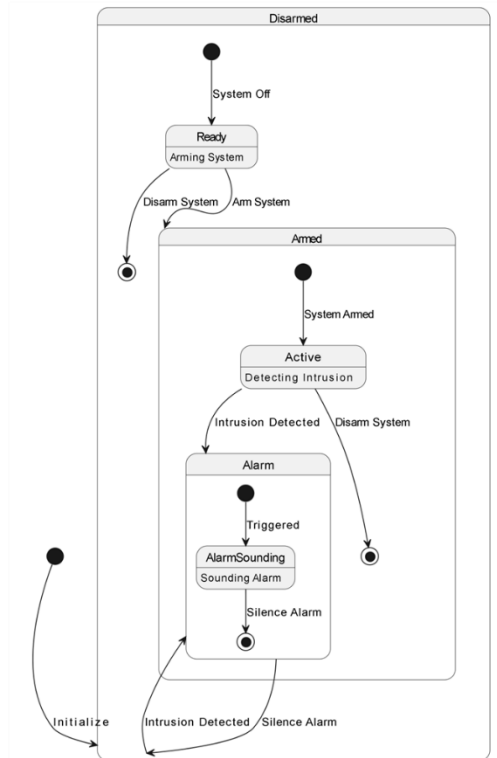
Class Diagram:

The class diagram of our system shows the necessary components and shows how we have allowed the capture data from the different sensors into the remote database to be processed



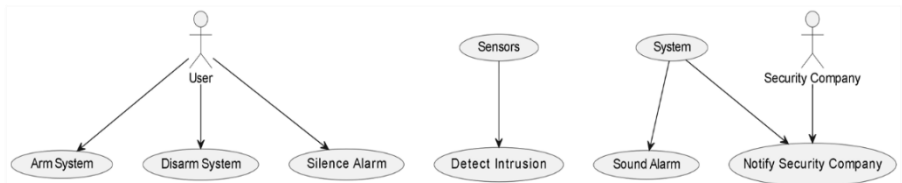
State Diagram

The sequence diagram shows cases how our home security system will work, from the system off scenario to the Armed scenario and what occurs within them.



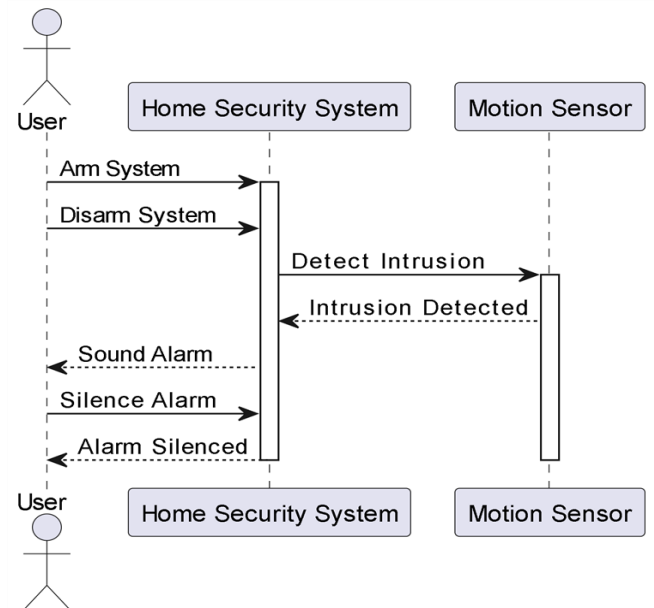
Case Diagram

The case diagram shows the overview of the system functionalities and the different entities involved in setting up the security system



Sequence Diagram

The Sequence diagram shows a specific use of the system. In this case it showcases how when motion is detected and the system is armed an alarm will be activated.



Code Implementation Process

The way we approached in determining the necessary components of the home security system is first evaluating what is needed in a home security system, we determined that the sensor of motion, temperature, humidity, and pressure were the most important sensors.

We then implemented a way to capture the data from the sensor hat and present the information. Once we were able to capture the data, we designed a table in the remote database and wrote up the code that allowed us to send the captured data into the table. We also implemented away for the system to be activated and deactivated, as well as when activated it is able to sound an alarm.

Contribution of Work

My team member was responsible with writing up the code that gathers data from each sensor,

Whereas I was responsible with implementing the code that will the data to sent to the remote database in a readable table. I was also responsible with designing the system as a whole, from what each UML Diagram should involve and show. I also drafted on the possibility of what is required in the future to implement our system.

We both we part of the requirement process, determining the functional and non-functional requirements of the system.

Project Reflection

The project has proven to be partially successful as to get the entire system working there is a bit of future work involved to bring it to the real world. When researching how different home security systems worked allowed us to narrow down the necessary functions of the system and was able design it promptly. We did come into some unforeseen issues, mostly regarding the code as it didn't always allow us to send the code to the database or show up in a table form. We will need to design a UI for users to access the database and get alerts.

